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QUARTERLY EXAMINATION - 2020

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BIOLOGY

12 - Std

Time : 3 hrs. Marks : 70

Bio – Botany

I **Choose the correct answer.** **8x1=8**

1. The size of Myosotis pollen grains is:
   (a) 10 micrometers (b) 20 micrometers
   (c) 200 micrometers (d) 2000 micrometers
2. Which Mendelian idea is depicted by a cross in which the F1 generation resembles both the parents.
   (a) Incomplete dominance (b) Law of Dominance
   (c) Inheritance of one gene (d) Co-dominance.
3. The genes controlling the seven pea characters studied by Mendel are known to be located on how many different chromosomes?
   (a) Seven (b) Six (c) Five (d) Four
4. Changing the codon AGC to AGA represents
   (a). Missense mutation (b). Nonsense mutation
   (c).Frameshift mutation (d).Deletion mutation.
5. EcoRI cleaves DNA at
   (a). AGGGTT (b). GTATATC (c).GAATTC (d). TATAGC
6. In which techniques Ethidium bromideis used?
   (a). Southern blotting (b). Western blotting
   (c). Polymerase Chain Reaction (d).Agarose gel electrophoresis
7. The time duration for sterilization process by using autoclave is ___minutes and the temperature is___
   (a) 10 to 30 minutes and 125°C (b) 15 to 30 minutes and 121°C
   (c) 15 to 20 minutes and 125°C (d) 10 to 20 minutes and 121°C
8. Solidifying agent used in plant tissue culture is
   (a) Nicotinic acid (b) Cobaltous chloride (c) EDTA (d) Agar

II **Answer any four questions.** **4x2=8**

9. Write about pollenkitt.
10. Write two reasons for the success of Mendel's hybridization experiments.
11. What are multiple alleles?
12. What is bioremediation?
13. What is cybrid?
14. Write about cryopreservation.

III **Answer any three questions. Question number 19 is compulsory. 3x3=9**

15. Write the functions of the tapetum.
16. Name the scientists who rediscovered Mendelian principles.
17. What are the key enzymes used in genetic engineering (biotechnology)?
18. Distinguish between somaclonal variation and gametoclonal variation.
19. What do you understand from the term pBR322?

IV. **Answer all questions.** **2 x 5 =10**

20. (a) Describe the structure of an ovule with a suitable diagram. ***(OR)***
    (b) Describe incomplete dominance in Mirabilisjalapa (Four-o'clock plant)
21. (a). Mention the applications of biotechnology. ***(OR)***
    (b). Explain the basic principles involved in plant tissue culture.

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I **Note : (i) Answer all the questions. (ii) Choose the most appropriate answer from the given four alternatives and write the option code and the corresponding answer.** **8x1=8**

1. **Match the following list-I with list-II and select the correct option:**

Column A (Contraceptive Type)	**Column B (Example)**
(A) Copper releasing IUD	(i) LNG-20
(B) Hormone releasing IUD	(ii) Lippes Loop IUD
(C) Non-medicated IUD	(iii) Saheli
(D) Mini pills	(iv) Multiload-375

a) A-(iv), B-(ii), C-(i), D-(iii) b) A-(iv), B-(i), C-(iii), D-(ii)
c) A-(i), B-(iv), C-(ii), D-(iii) d) A-(iv), B-(i), C-(ii), D-(iii)

2. Sudden appearance of vestigial organs in highly evolved organisms is called
a) Vestigial organs. b) Homologous organs
c) Atavistic organs d) Analogous organs

3. Meselson and Stahl's experiment proved
a) Transduction b) Transformation
c) DNA is the genetic material d) Semi-conservative nature of DNA replication

4. Which of the following is incorrect regarding ZW-ZZ type of sex determination?
a) It occurs in birds and some reptiles.
b) Females are homogametic and males are heterogametic.
c) Males produce same type of gametes.
d) It occurs in gypsy moth.

5. In this type, only females are produced by parthenogenesis:
a) Arrhenotoky b) Amphitoky c) Thelytoky d) Strobilation

6. B cells that produce and release large amounts of antibody are called _____.
a) Memory cells b) Basophils c) Plasma cells d) Killer cells

7. The process which the sperm undergoes before penetrating the ovum is .....
a) Spermiation b) Cortical reaction c) Spermiogenesis d) Capacitation

8. Amphetamines are stimulants of the CNS, whereas barbiturates are _____.
a) CNS stimulant b) Hallucinogenic c) Both a and b d) CNS depressants

II **Note : Answer any four of the following questions.** 4 x 2 = 8

9. What are operons? How many operon groups are present in *E. coli*?
10. Expand the following terms: a) ZIFT b) ICSI
11. What is known as the "Let-Down" reflex?
12. Write any TWO differences between a normal cell and a cancer cell.
13. What is Plasmotomy?
14. List out the major gases thought to be present in primitive earth.

III **Note : Answer any three of the following questions.** **3 x 3 = 9**
Question number 19 is compulsory.

15. What is Cooley's anemia?
16. List out the chemical alarm signals produced during inflammation.
17. Mention any three objections to Darwinism.
18. Draw a neat labeled diagram of the human ovary structure.
19. What is Kornberg enzyme? State its significance.

IV **Note : Answer all the questions in detail.** **2 × 5 = 10**

20. a) What are the applications of Karyotyping? ***(OR)***
b) Explain the structure of Immunoglobulin (Ig) with a suitable neat labeled diagram.

21. a) Explain the various phases of the human menstrual cycle. ***(OR)***
b) Explain the evolutionary path and lineage of modern man.

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